

UGANDA ADVANCED CERTIFICATE OF EDUCATION

PHYSICS PAPER 1 SET 2 2026

2 HOURS 30 MINUTES

INSTRUCTIONS

This paper consists of four sections A, B, C and D.

This paper contains 7 scenario items and a learner answers only 4 items.

Section A has **one compulsory** item and a candidate must answer **one item** from each of sections B,C, and D.

All responses must be written on answer booklets provided.

Any extra item attempted will not be scored

SECTION A (PARTICLE EMISSION)

Item one

Uranium is one of the most valuable minerals because of it is used to generate energy in nuclear reactors. Uranium has two isotopes that is; ${}^{235}_{92}\text{U}$ and ${}^{238}_{92}\text{U}$ and the most commonly used isotope is ${}^{235}_{92}\text{U}$. So many governments want to shift from other sources of energy to nuclear energy.

${}^{235}_{92}\text{U} + {}^1_0\text{n} \rightarrow {}^{95}_{42}\text{Mo} + {}^{139}_{35}\text{La} + 2{}^1_0\text{n} + 7{}^0_{-1}\text{e}$ is one of the typical equations that can be used to generate energy. Uganda nuclear energy committee was seeking prior knowledge on the generation of electricity by the process and its related effects in short run and long run. A radiographer got a small volume of solution which contained a radioactive isotope of sodium and an activity of 2000 disintegrations per minute which he injected into the blood stream of patient. After 30 hours the activity of 1.0 cm^3 of the blood was found to be 0.050 disintegrations per minute. If the half-life of the sodium isotope is taken as 15 hours, A Radiographer had to determine the volume of blood in the patient in order to guide him properly. A normal adult should have above 10 liters of blood.

Task: address the concerns of the radiographer and the Uganda nuclear committee.

SECTION B (FORCES AND MOTION)

Item two

A technician uses a 5 meter long wooden ladder to reach a security light mounted on a wall of a building. the ladder leans against a smooth wall, making an angle of 60° with the ground. The base of the ladder rests on a rough concrete surface to prevent slipping. The ladder has a mass of 20 kg and is uniform. As the technician of mass 60 kg climbs the ladder, there is a limit to how high he can go before the ladder begins to slip. The technician doesn't know how to determine how high he should climb before the ladder begins slip.

A red train travelling at 72 kmhr^{-1} and a green train travelling at 144 kmhr^{-1} are headed toward each other along a straight , level track, when they are 950 m apart, each engineer sees the others train and applies the brakes. The brakes slow each train at a rate of 1.0 ms^{-2} . there was a need to determine whether there was collision inorder to obtain the speed of red train and speed of green train after impact, and if no collision to determine the separation between trains when they stop.

Item three

A baseball is hit at Fenway Park in Boston at a point 0.762 m above the home plate with an initial velocity of 33.53ms^{-1} directed 55.0° above the horizontal. The ball is observed to clear the 11.28 m high wall in left field 5.00 seconds after it is hit at a point just inside the left field foul line pole. There was a need to determine the horizontal distance down the left field foul line from home plate to the wall, the vertical distance by which the ball clears the wall, the horizontal and vertical displacements of the ball with respect to home plate 0.500 seconds before it clears wall.

A house is built on the top of a hill with a nearby slope at an angle of 45° to the horizontal. an engineering study indicates that the slope angle should be reduced because the top layers of soil along slope might slip past the lower layers. If the coefficient of friction between two such layers is 0.5. the engineering are seeking the least angle through which the present slope should be reduced to prevent slippery.

A car of mass 1 tones tows a caravan of mass 600kg up a road which rises 1 m vertically for every 20 m of length. There are constant frictional resistances of 200 N and 100 N to the motion of car and that of caravan respectively. The combination has an acceleration of 1.2ms^{-2} with the engine exerting a constant driving force which shouldn't exceed 2KN.

There was a need to determine the tension in the towing bar and driving force.

Task: address the challenges in the scenarios above

SECTION C (ENERGY)

Item four

A certain factory has machines which work on a daily basis and it is noticed that due to that, they have been complaining about machines becoming too hot every day due to over work. One of the workers suggested that they should introduce in a liquid that will always cool their machines and must be of high specific heat capacity. They were presented with Liquid X and water and wished to know which was best to be used. The worker decided to use continuous flow experiment to find out the best. While using liquid X, It was found that when a p.d applied was 12.0V, current 1.5A, a rate of flow of liquid of 50.0g/minute cause the temperature of inflow liquid to differ by 10°C . When the p.d was increased to 16.0V with current of 1.6A, the rate of flow of 90.0g/minute was required to produce the same temperature difference as before. The factory manager had less knowledge to interpret all this for the best choice.

In solar water heater, the energy from the sun is gathered by water that circulates through tubes in a rooftop collector. The solar radiation enters the collector through a transparent cover and warms the water in the tubes, this water is pumped into a holding tank. Assume that the efficiency of the overall system is 20% that is 80% of the incident solar energy is lost from the system. What collector area is necessary to raise the temperature of 200 liters of water in the tank from 20°C to 40°C in 2.0 hours when the intensity of incident sunlight is 700Wm^{-2} .

Task : address the challenges in the above scenarios in order to give clear recommendations

Item five

- (a) During sporting events within large, densely packed stadiums spectators will send a pulse around the stadium. As the wave reaches a group of spectators, they stand with a cheer and then sit. At any instant the width of wave is the distance from leading edge (people just about to stand) to trailing edge (people have just sat down). Suppose a human wave travels a distance

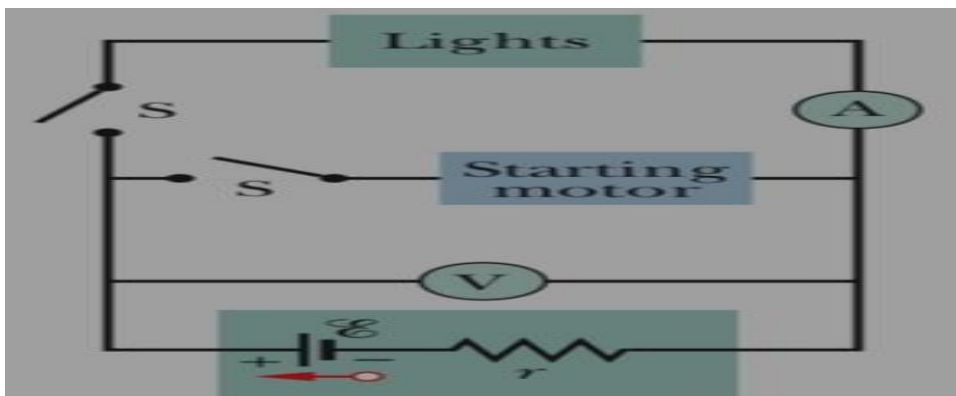
of 853 seats around a stadium in 39 seconds, with spectators requiring about 1.8 seconds to respond to the wave's passage by standing and sitting. Determine wave speed in seats per second and width in number of seats.

- (b) At Kasaana Clinic, the solar-powered mirror galvanometer used for detecting weak currents in patients' ECG sensors gives unstable readings. The technician suspects poor mirror alignment and wrong mirror size. Nearby, the clinic's corridor security. Uses two inclined mirrors at 90° , but staff complain the images deviate confusingly. Also, some short patients, averaging 1.50 m tall, can't see their full reflection in the mirror above the hand washing station. There was a need to help the clinic to determine correct alignment of mirrors and help the technician to know the deviation angle between two mirrors and minimum mirror height he should install for full self-view
Address the challenges of the technician

SECTION D (FIELDS AND CHARGES)

Item 6

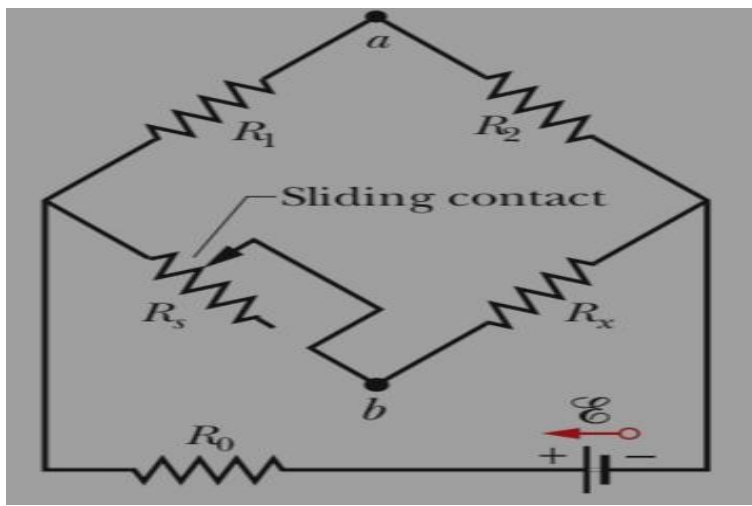
- (a) A 10 km long underground cable extends east to west and consists of two parallel wires each of which has resistance $13\Omega/\text{km}$. An electrical short develops at distance x from the west end a conducting path of resistance R connects the wires. The resistance of the wires and short is then 100Ω when measured from the east end and 200Ω when measured from the west end. Determine the values of x and R .
- (b) When the lights of the car are switched on, an ammeter in series with them reads 10.0 A and a voltmeter connected across them reads 12.0 V. when the electric starting motor is turned on, the ammeter reading drops to 8.00 A and the lights dim somewhat. If the internal resistance of the battery is 0.0500Ω and that of the ammeter is negligible. Determine the emf of the battery and current through the starting motor when the lights are on?



- (c) Using the support material below R_x is to be adjusted in value by moving the sliding contact across it until points a and b are brought to the same potential. One tests for this condition by momentarily connecting a sensitive ammeter between a and b; if these points are at the same potential, the ammeter will not deflect. Show that when this adjustment is made, the following relation

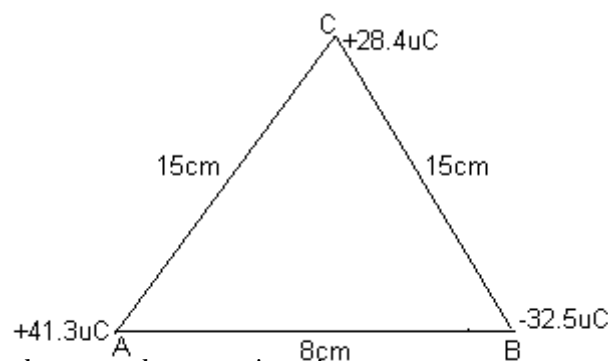
$$R_x = \frac{R_s R_2}{R_1}$$

An unknown resistance R_x can be measured in terms of a standard resistor R_s using this device which is a Wheatstone bridge.



item 7

In one of the tests done by the engineer using an electronic microscope it was found that three paint particle from the nozzle of the spray were at one time arranged in a shape of a triangle ABC and carrying charge as shown in the figure below.



There was a need to know how an electrostatic paint spray works and determine the electrostatic force experienced by the charge at point C

As part of a national science initiative, a group of O-Level students visited a newly built government research center. One of the highlights of the visit was a demonstration of a Van de Graaff generator, a machine that builds up very high voltages. The students were curious and wondered how the generator could reach such high potentials. The operator explained that the large metal sphere on top acted as an equipotential surface. He added that the sphere needed high capacitance to store large amounts of charge, but warned that the electric field intensity around it must not exceed 30,000 volts per centimeter. If it did, the air would break down and a spark would jump out.

They also asked what an equipotential surface was, what the maximum potential of the sphere could be before a breakdown, and what its capacitance would be at that potential.

Later in the demonstration, each student was given a small metal piece to hold. As they moved closer to the Van de Graaff sphere, they felt a pulling force toward it. The force grew stronger the closer they got, and this puzzled them.

Learners wanted to understand the structure and mode of operation of the generator and nature of the spherical surface of the generator

Task : address the needs in the above scenarios

